

SLIDE 1: TITLE PAGE

Problem Statement ID: PS-CYBER-01

Problem Statement Title: AI-Powered Behavioral Anomaly Detection for Cybersecurity

Theme: Cyber Security / Artificial Intelligence & Machine Learning

PS Category: Software

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SLIDE 2: PROPOSED SOLUTION

CyberTwin AI: Autonomous Threat Fusion & Behavioral Digital Twins

Detailed Explanation of Proposed Solution:

- **Autonomous Cyber Digital Twin Platform:** Continuous threat detection engine learning baseline normal behavior across access logs, network topologies, and command sequences.
- **Tri-Model Consensus Defense:** Combines PyTorch Autoencoders (point-in-time anomalies), Sequence Transformers (temporal attacks), and In-Memory NetworkX Graph Topology (lateral movement).
- **Pre-Emptive Trajectory Prediction:** Autoregressive Behavioral Twin model to mathematically forecast attacker next moves $P(a_{t+1}|a_1..a_t)$ before execution.

How it Addresses the Problem:

- **Zero-Rule Dependency:** Replaces static signatures with behavioral baseline modeling to catch novel zero-days.
- **Extreme Class Imbalance Mitigation:** SMOTE oversampling & Focal Loss (pos_weight=10.0) for rare intrusions (<1% traffic).
- **Cold-Start & Drift Handling:** K-Means Baseline Clustering & rolling temporal windows.

Innovation & Uniqueness:

- **Predictive Defense:** Forecasts next attacker actions (82% probability DB exfiltration) rather than reactive logs.
- **XAI + NLG:** Translates mathematical SHAP feature attributions into plain English incident reports.
- **SOAR Playbooks:** One-click automated remediation (Force MFA, Host Isolation, Dynamic Decoys).

SLIDE 3: TECHNICAL APPROACH

Technologies Used & System Architecture Flow

- **AI/ML & Math:** PyTorch 2.x, XGBoost, SHAP, NetworkX, Scikit-learn, NumPy, Pandas.
- **Backend & Streaming:** Python 3.12, FastAPI, Async WebSockets, Redis Streams (Pub/Sub & Feature Store), PostgreSQL.
- **Frontend Dashboard:** React 19, TypeScript, Vite, Tailwind CSS, Zustand, Recharts, Lucide Icons.
- **DevOps & MLOps:** Docker (Multi-stage CPU optimization), GitHub Actions (CI/CD), Render IaC (render.yaml).

```
[ Synthetic Telemetry Stream ] --> [ FastAPI Ingestion / Redis Streams ]  
--> [ PyTorch Autoencoder | Sequence Transformer | Graph Topology Engine ]  
--> [ Anomaly Fusion Engine ] (Master Risk Score = Fusion x Consensus Boost)  
--> [ Multi-Class XGBoost ATT&CK; & Autoregressive Twin Predictor ]  
--> [ SHAP XAI Engine + NLG Translator ] --> [ React SOC War Room & SOAR Playbooks ]
```

SLIDE 4: FEASIBILITY AND VIABILITY

Feasibility, Risk Mitigation & Performance Strategies

Analysis of Feasibility:

- **High Engineering Feasibility:** Functional microservices architecture deployed with real-time WebSockets (<50ms latency).
- **Enterprise Scalability:** Async Redis consumer groups distribute ML inference across background workers.
- **Domain Agnostic:** Scales across enterprise IT infrastructure, IoT edge hubs, POS terminals, and OT networks.

Strategies for Overcoming Challenges:

- **Dynamic Consensus Multipliers:** Strict top 1% alert budget threshold (Risk > 0.75) with tri-model agreement multipliers to suppress false positives.
- **Peer-Role Baseline Clustering:** Cold-start entities temporarily inherit baseline profiles of peer roles until 100+ events are logged.
- **Lightweight CPU Quantization:** PyTorch CPU-optimized inference images (<800MB) execute inference in under 30ms.

SLIDE 5: ARTIFACTS

System Code Architecture & Application Dashboards

```
cybertwin-ai/  
|-- backend/ (FastAPI, Schemas, Anomaly Fusion, SHAP XAI, SOAR Playbooks)  
|-- behavior_twin/ (Autoregressive LSTM Predictor, Token Mapper)  
|-- graph_engine/ (NetworkX Lateral Movement Graph Manager)  
|-- ml/ (PyTorch Autoencoder, Sequence Transformer, XGBoost Classifier)  
|-- dashboard/ (React 19, TypeScript, Tailwind, Zustand War Room UI & Click Simulator)  
-- docker/ & render.yaml (Multi-stage Containers & IaC Cloud Blueprint)
```

Key Screenshots Verified & Included:

1. **SOC Executive Overview:** Real-Time Risk Score Timeline, 2,561 Analyzed Events, 500 Digital Twins, 99.4% Consensus Score.
2. **Live Threat Feed & Simulator:** Real-time severity-ranked alerts (Impossible Travel - 98.0% Critical, Lateral Movement - 84.0% High) with Top Click Simulator.
3. **Threat War Room & XAI:** Natural Language Incident Report, SHAP Risk Feature Attributions (velocity_kmh: +48.0%), 3-Hop Graph Topology, Behavioral Twin Forecast.

SLIDE 6: RESEARCH AND REFERENCES

Reference Frameworks & Live Demonstration Endpoints

Reference Work & Academic Basis:

- **MITRE ATT&CK; Framework:** Threat modeling and attack taxonomy mapping. (<https://attack.mitre.org/>)
- **SHAP (SHapley Additive exPlanations):** Lundberg & Lee (2017), NeurIPS.
- **Attention Is All You Need:** Vaswani et al. (2017), Causal Transformer architecture.
- **Network Topology & Graph Mining:** NetworkX Directed Multigraph algorithms.

Live Working Demonstration:

- **Frontend Dashboard:** <http://localhost:5173> (Production Vite Build with Ingest Simulator)
- **API & Interactive Swagger Docs:** <http://localhost:8000/docs>
- **Render Cloud Deployment:** <https://cybertwin-dashboard.onrender.com>
- **GitHub Monorepo:** [justynigam/AI-Cybertwin](https://github.com/justynigam/AI-Cybertwin)